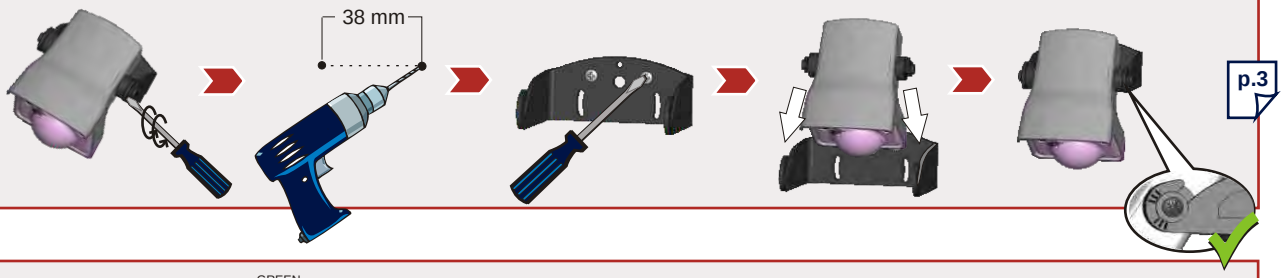
OPEN UP NEW HORIZONS

CONDOR
QUICK GUIDE

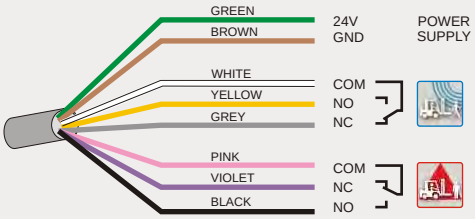
MOTION & PRESENCE SENSOR FOR AUTOMATIC INDUSTRIAL DOORS

STEP 1 MOUNTING



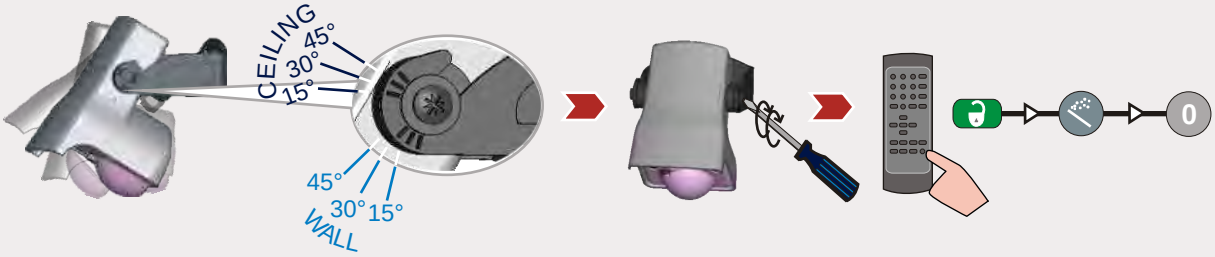
p.3

STEP 2 WIRING



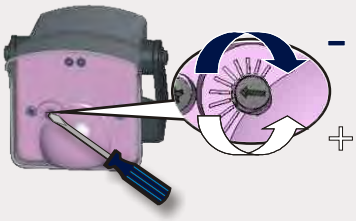
p.3

STEP 3 SENSOR ANGLE



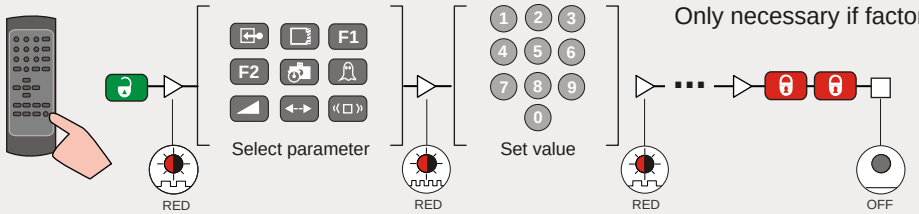
p.4

STEP 4 RADAR ANGLE



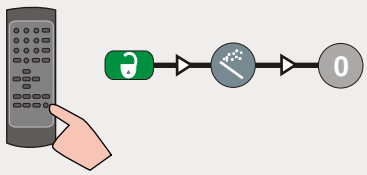
p.4

STEP 5 RC SETTINGS



p.5

STEP 6 SETUP

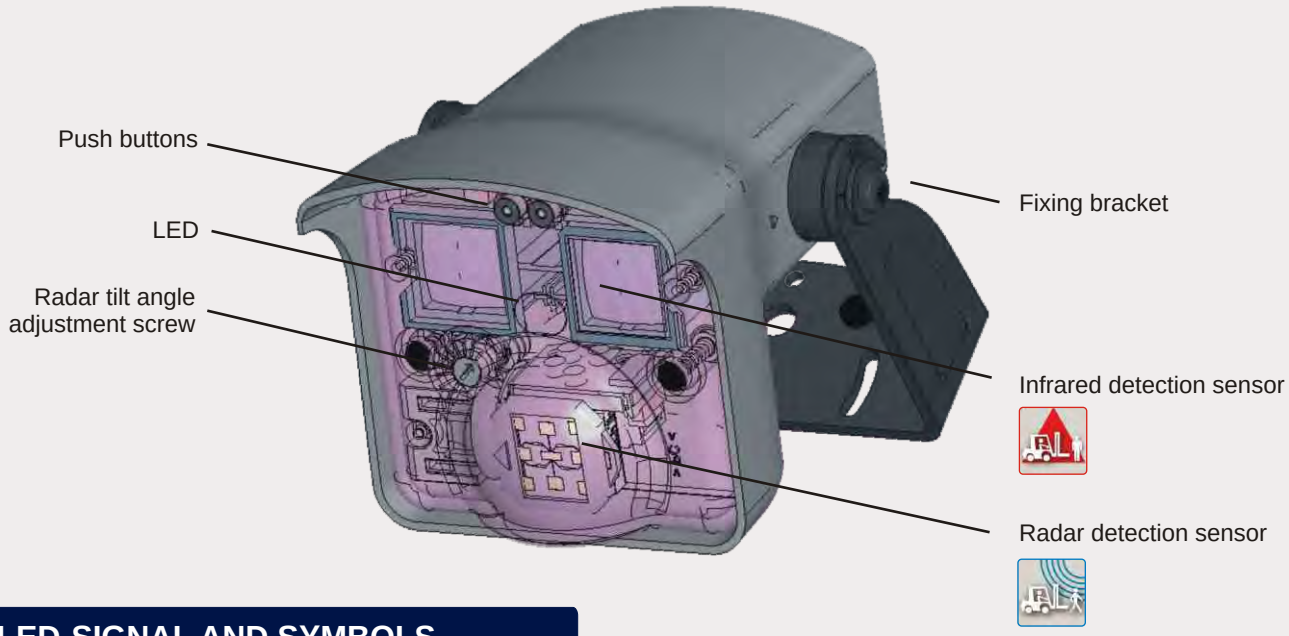


See user's guide on the next pages for more detailed information.

p.6

MOTION & PRESENCE SENSOR FOR AUTOMATIC INDUSTRIAL DOORS

DESCRIPTION



LED-SIGNAL AND SYMBOLS

GREEN LED
RADAR

RED LED
IR

RED-GREEN LED
SETUP

LED
flashes quickly

LED
flashes slowly

LED
OFF

MOTION

PRESENCE

FACTORY
VALUE

p.2
SEE PAGE 2

IMPORTANT!

INSTALLATION TIPS



Fasten the sensor firmly in order to avoid vibrations.



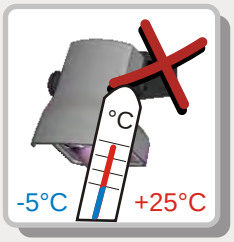
Do not cover the sensor.



Avoid moving objects in front of the sensor



Avoid high intensity lightning in the detection field.

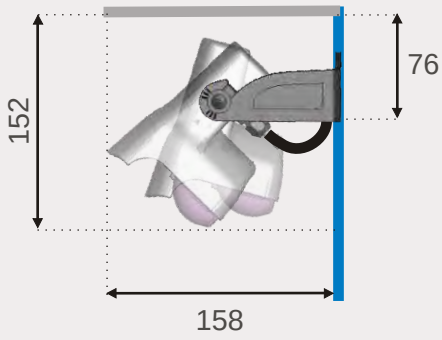


Avoid exposing the sensor to sudden and extreme temperature changes.

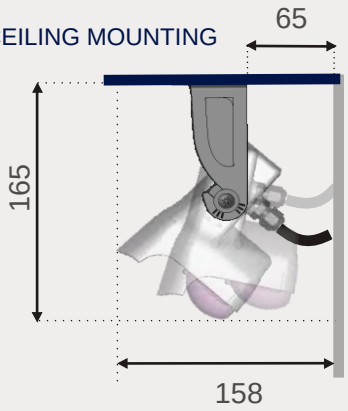
STEP 1 MOUNTING

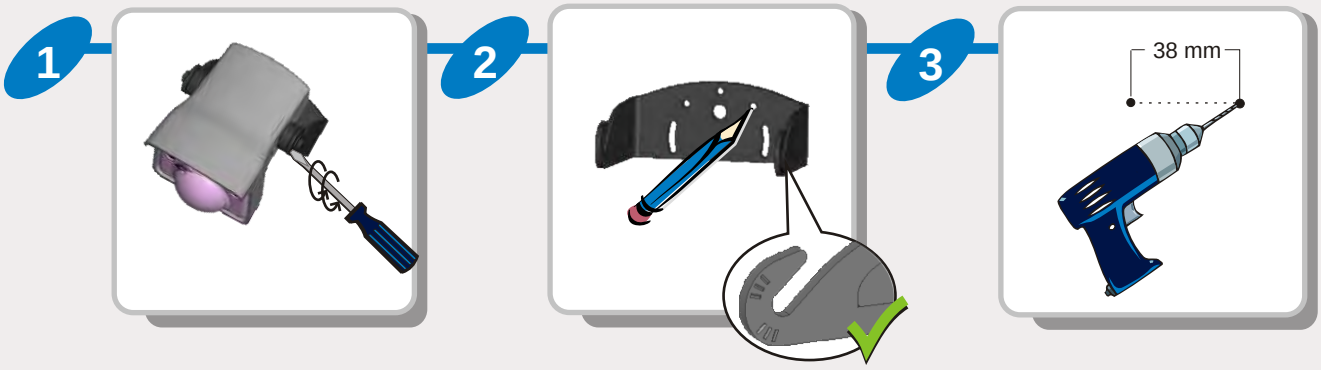
Minimum distances in mm required to fully adjust the sensor:

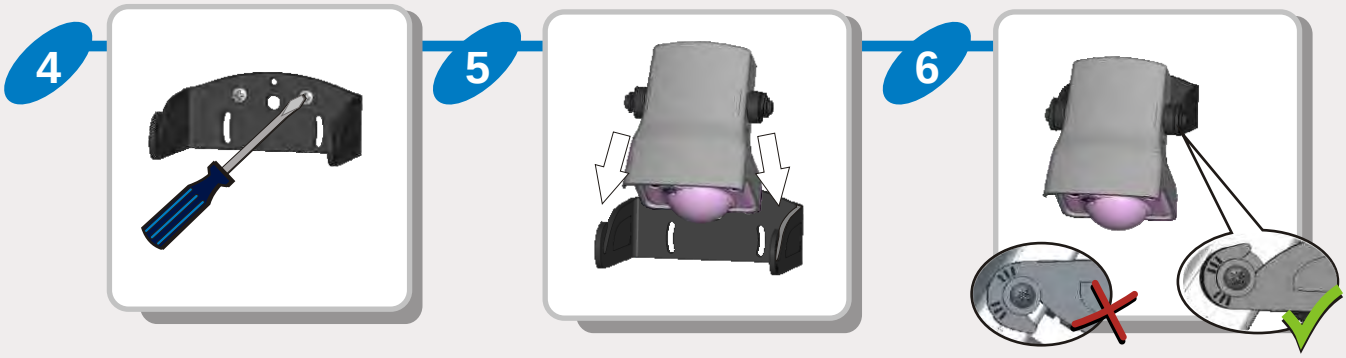
WALL MOUNTING



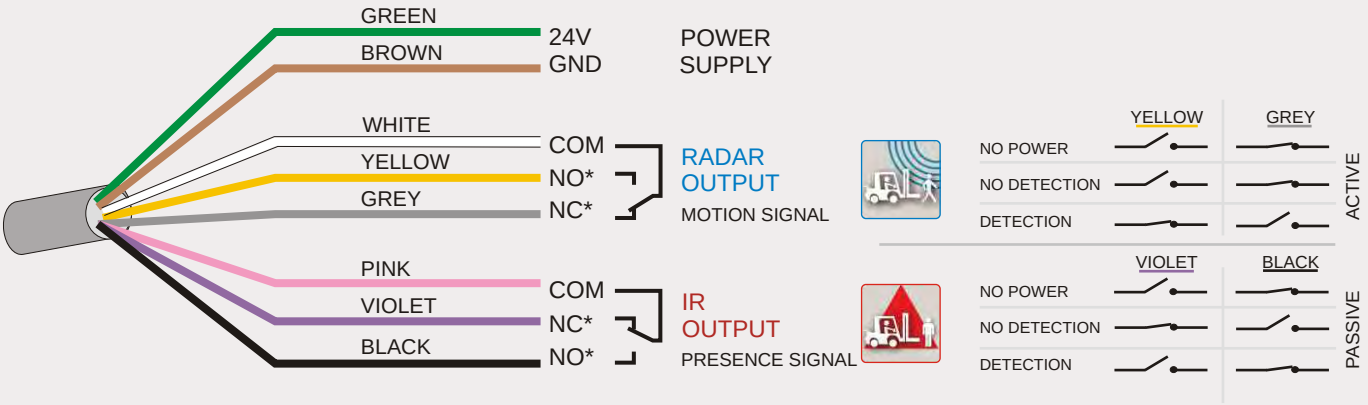
CEILING MOUNTING







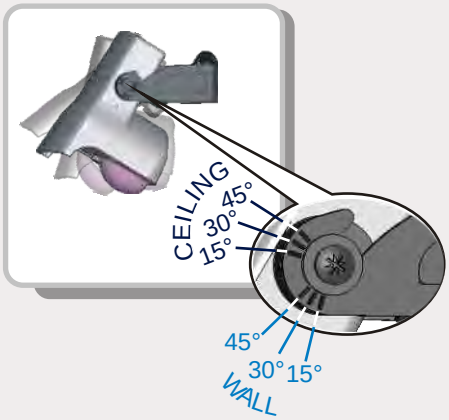
STEP 2 WIRING



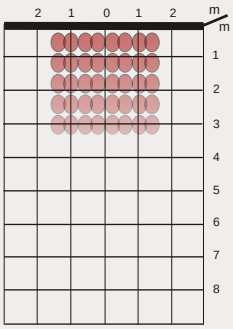
*Powered, no detection

ANGLE ADJUSTMENTS

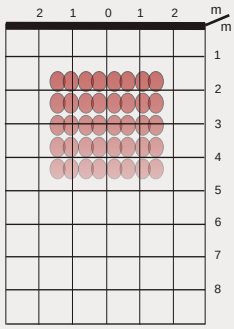
STEP 3 SENSOR ANGLE



RECOMMENDED

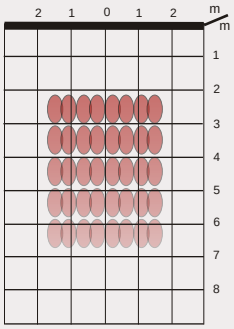


Sensor angle: 20°
Mounting height: 5m

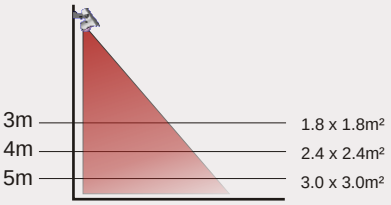


Sensor angle: 30°
Mounting height: 5m

NOT RECOMMENDED

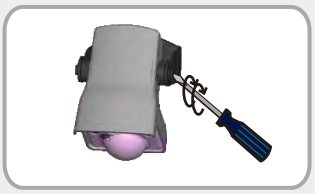


Sensor angle: 45°
Mounting height: 5m

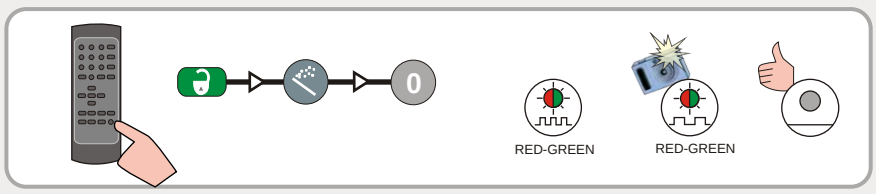


Guideline values for a 50 x 50cm² testbody.

The spotfinder can detect a slightly bigger zone.

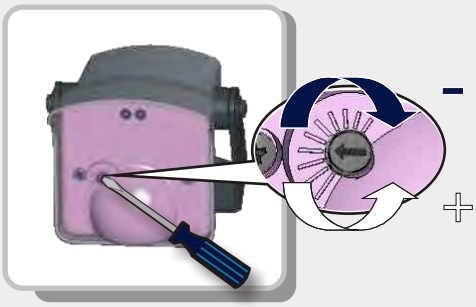


Tighten the screws firmly.



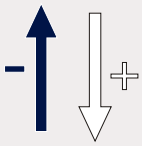
Launch a setup to make a reference picture.
Step out of the detection field and do not leave any tools inside the detection field during the setup procedure.

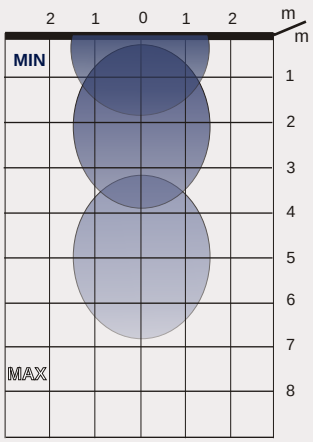
STEP 4 RADAR FIELD ANGLE



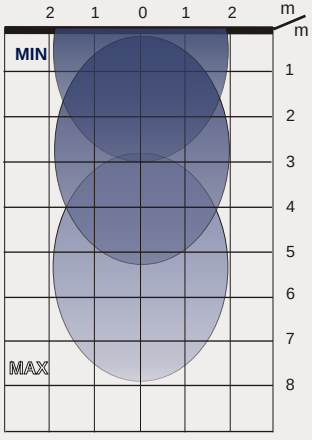
\- By turning the tilt angle adjustmentscrew **clockwise**, the radar field angle is **reduced**.

\+ By turning the tilt angle adjustmentscrew **counter clockwise**, the radar field angle is **increased**.





Mounting height: 3.5m
Sensor angle: 20°
Sensitivity: 9



Mounting height: 5m
Sensor angle: 20°
Sensitivity: 9

STEP 5 SETTINGS

Only necessary if factory values do not match.

unlock

lock

value check

setup

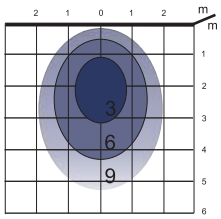
Output redirection F1

	Radar output	IR output
0	motion	presence
5	airlock mode	presence

Sensitivity

0-9

Factory value: 7



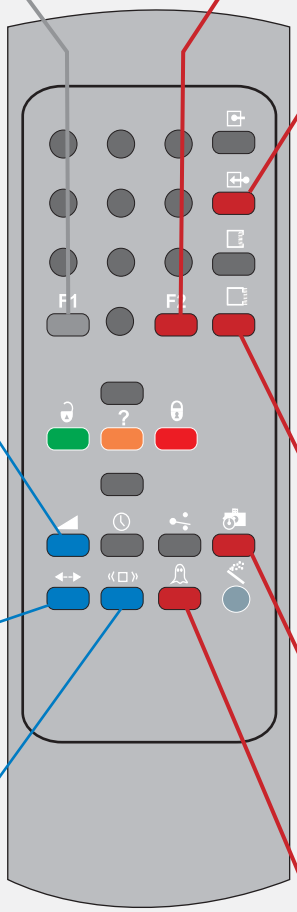
Mounting height: 5m
Sensor angle: 20°
Radar field angle: 30°

Detection mode

1 bidirectional
2 unidirectional
3 unidirectional reverse

Rejection mode

1 detection of all targets in motion
2 all targets in motion + enhanced immunity
3 low pedestrian/parallel traffic rejection
4 medium pedestrian/parallel traffic rejection
5 high pedestrian/parallel traffic rejection



Minimum size of target F2

1 2 3 4 5 6 7

The position of the target in the field is random.

IR-detection field

1 2 3 4 5 6 7 8 9

Pulse frequency

1 frequency 1
2 frequency 2

Max. duration of presence detection

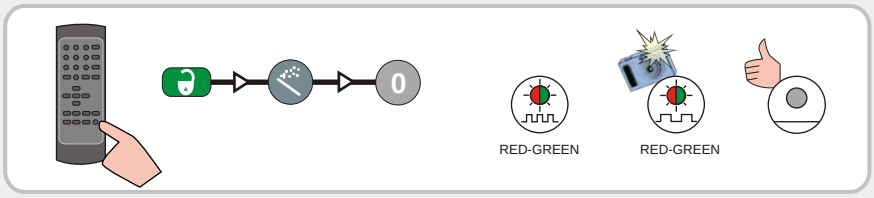
0	30s	5	20min
1	1min	6	1h
2	2min	7	1h30
3	5min	**8**	**2h**
4	10min	9	∞*

* not guaranteed

IR-curtain immunity

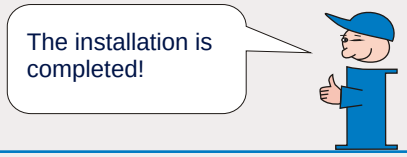
1 reduced
2 standard
3 increased

STEP 6 SETUP



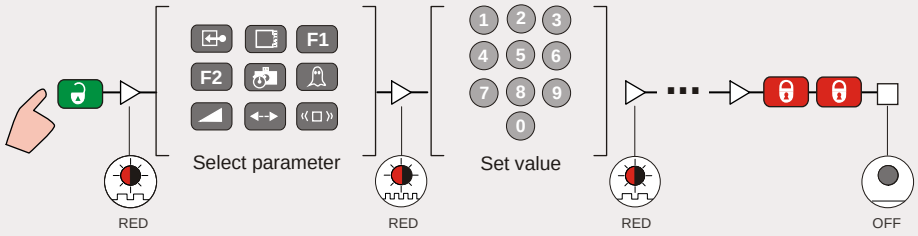
Launch a setup to make a reference picture.

Step out of the detection field and do not leave any tools inside the detection field during the setup procedure.

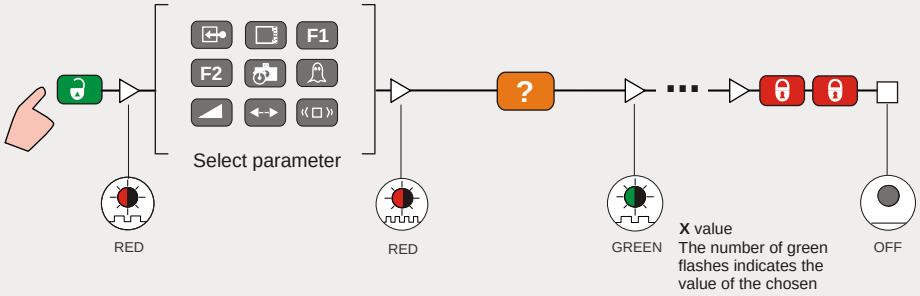


HOW TO USE?

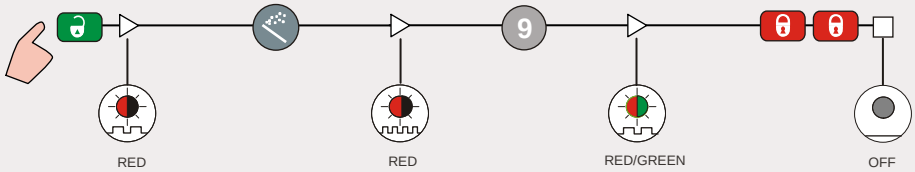
How to adjust one or more parameters?



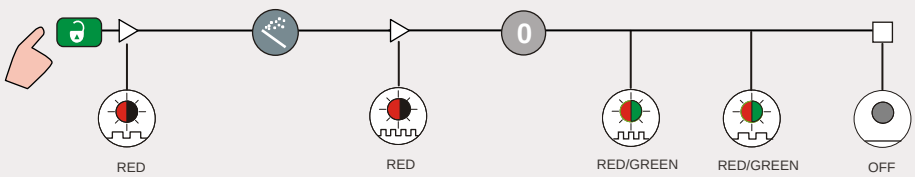
How to check one or more values?



How to reset to factory values?



How to launch a setup?



TROUBLESHOOTING

SYMPTOMS	LED	POSSIBLE CAUSES	CORRECTIVE ACTION
The red LED lights up.	 RED	The sensor detects a presence.	Launch a setup. p.6
			Wait as long as the time set in the “maximum duration of presence detection” setting. p.5
The red LED is permanently on after a setup.	 RED	The sensor fails the IR setup.	Cut and restore the power supply and launch a new setup (please step out of the IR field). p.6
The red LED is on.	 RED	The presence detection is disturbed by the rain.	Increase the immunity of the IR field (value 3). p.5
	 RED	The sensor is not installed properly.	Fasten the sensor firmly. p.4
The red LED flashes quickly.	 RED	Two or more sensors are installed in proximity to each other and cause interferences.	Select a different frequency for each sensor. p.5
The IR sensor does not react.	 RED OFF	The IR power emission is too low according to the mounting height.	Launch a new setup. Make sure to step out of the detection field before. p.6
The door keeps recycling open-closed.	 RED	The sensor is disturbed by the door motion, because it sees the door or feels vibrations.	Increase the sensor angle. p.4
	 GREEN		Increase the radar field angle and the level of rejection mode. p.4-5
The green LED is on.	 GREEN	The radar detection is disturbed by the rain.	Check whether the unidirectional mode is selected. p.5
			Increase the level of rejection mode. p.5
The sensor detects objects outside of its detection field.	 GREEN	Too much reflection due to a metallic environment.	Increase the level of rejection mode. p.5
The sensor does not respond to the remote control.		The batteries in the remote control are not installed properly or dead.	Check or replace the batteries.
		The remote control is badly pointed.	Point the remote control towards the sensor.
		The sensor is not powered.	Check the power supply of the sensor.
		The sensor requires an access code.	Cut and restore power supply. No code is required during the first minute after powering.
The sensor does not unlock, even after entering the access code.	 RED	You have to enter a code or the wrong code was entered. (Maybe it was changed?).	New access code: + + new code Delete access code: + + 0000

TECHNICAL SPECIFICATIONS

Supply voltage:	12 - 24V AC +/- 10% 12 - 24V DC +10% / - 3%
Mains frequency:	50 - 60 Hz
Power consumption:	< 3.5W / VA
Output:	2 relays with switch-over contact (voltage free)
Max. Voltage:	42V AC / DC
Max. Current:	1A (resistive)
Max. switching power:	30W (DC) / 48VA (AC)
Installation height:	2.5m - 6m*
Temperature range:	-30°C to +60°C
Protection degree:	IP65
Humidity:	0-95% non-condensing
Norm conformity:	Electromagnetic compatibility (EMC): 2004/108/EEC, R&TTE: 1999/5/EC
Dimensions:	127mm (D) x 102mm (W) x 96mm (H)
Housing material:	ABS
Front face material:	polycarbonate
Housing colour:	black
Front face colour:	smoked violet
Cable length:	10m

		
Technology:	microwave	infrared
Radiated frequency/wavelength:	24.150 GHz	875nm
Radiated power density	< 5mW/cm^2	< 250mW/m^2
Detection type:	motion	motion & presence
Detection field:	4m x 5m**	3m x 3m**
Output hold time:	0.5s - 9s (adjustable)	0.5s
Reaction time:	100ms	250ms
Min. target speed:	5cm/s (in sensor axis)	5cm/s to activate a detection
Tilt angle:	-8° to 22° (relative to sensor front face)	15°- 45°
LED-signal:	green	red

*depending on size and nature of target
**mounting height: 5m
Specifications are subject to changes without prior notice.

EU declaration of conformity

BEA hereby declares that the product CONDOR is in conformity with the basic requirements and the other relevant provisions of the standards 1999/5/EC and 2004/108/EEC.

The complete declaration of conformity is available on our website: www.bea.be

The colour version of this user's guide and other technical documentation are available on our website (www.bea.be) or can be requested by phone (+32 4 361 65 65) or mail (info@bea.be).

BEA SA - LIEGE Science Park - Allée des Noisetiers 5 - B-4031 Angleur - T +32 4 3616565 - F +32 4 3612858 - info@bea.be - www.bea.be